

Erdős Problem #575

EP-575	L1 - Tractable	Ouvert	Graph Theory
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Énoncé original (anglais, reproduction mot pour mot)

If $\mathcal{F}$ is a finite set of finite graphs then $\mathrm{ex}(n; \mathcal{F})$ is the maximum number of edges a graph on n vertices can have without containing any subgraphs from $\mathcal{F}$. Note that it is trivial that $\mathrm{ex}(n; \mathcal{F}) \leq \mathrm{ex}(n; G)$ for every $G \in \mathcal{F}$.

Is it true that, for every $\mathcal{F}$, if there is a bipartite graph in $\mathcal{F}$ then there exists some bipartite $G \in \mathcal{F}$ such that

$\mathrm{ex}(n; G) \sim \mathrm{ex}(n; \mathcal{F})$?

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